Molecular Characterization Laboratory
Frederick National Laboratory for Cancer Research
(Leidos Biomedical Research, Inc.)
459 Miller Drive, Frederick, MD 21702
T: 301.846.7689

Patient Name:		Patient DOB:	
Date Collected:		Referring Physician:	
Biopsy Site:	**Bone Marrow**	Primary Diagnosis:	**Acute Myeloid Leukemia**
Sample ID:		MoCha ID:	
Telephone:		Fax:	
Tumor Content (%):	**100%**		

Preanalytics to include tumor enrichment as required is performed by Roswell Park Comprehensive Cancer Center; Buffalo; NY 14203

Relevant Acute Myeloid Leukemia Findings

Gene	Finding	Gene	Finding
ABL1	None detected	MECOM	None detected
ASXL1	None detected	MLLT3	None detected
CEBPA	None detected	MYH11	None detected
CREBBP	None detected	NPM1	None detected
FLT3	***FLT3 ITD mutation***	NUP214	None detected
IDH1	None detected	RARA	None detected
IDH2	None detected	RUNX1	None detected
KMT2A	None detected	TP53	None detected

Relevant Biomarkers

Tier	Genomic Alteration	Relevant Therapies (In this cancer type)	Relevant Therapies (In other cancer type)	Clinical Trials
IA	*FLT3 ITD mutation*	**gilteritinib** [1] **midostaurin + chemotherapy** [1] azacitidine cytarabine + daunorubicin cytarabine + daunorubicin + etoposide cytarabine + etoposide + idarubicin cytarabine + fludarabine + idarubicin + filgrastim cytarabine + idarubicin cytarabine + mitoxantrone decitabine sorafenib sorafenib + chemotherapy venetoclax + chemotherapy	None	31

Prognostic significance: ELN 2017: Adverse

Public data sources included in relevant therapies: FDA1, NCCN
Public data sources included in prognostic and diagnostic significance: NCCN
Tier Reference: Li et al. *Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists.* J Mol Diagn. 2017 Jan;19(1):4-23.

Relevant Biomarkers (continued)

Tier	Genomic Alteration	Relevant Therapies (In this cancer type)	Relevant Therapies (In other cancer type)	Clinical Trials
IIC	*NUP98-NSD1 fusion*	None	None	2

Public data sources included in relevant therapies: FDA1, NCCN
Public data sources included in prognostic and diagnostic significance: NCCN
Tier Reference: Li et al. *Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists.* J Mol Diagn. 2017 Jan;19(1):4-23.

Variant Details

DNA Sequence Variants

Gene	Amino Acid Change	Coding	Variant ID	Locus	Allele Frequency	Transcript	Variant Effect
FLT3	p.(E608_N609insGYVDFREYEYDLKWEFPRE)	c.1823_1824insAGGCTACGTTGATTTCAGAGAATATGAATATGATCTCAAATGGGAGTTTCCAAGAGA	.	chr13:28608232	8.30%	NM_004119.3	nonframeshift Insertion
FLT3	p.(E596_Y597insEGGGVP)	c.1788_1788delAinsAGAGGGAGGGGGAGTCCCT	.	chr13:28608268	6.20%	NM_004119.3	nonframeshift Block Substitution

Gene Fusions (RNA)

Genes	Variant ID	Locus
NUP98-NSD1	NUP98-NSD1.N12N6	chr11:3765739 - chr5:176662822

Low Coverage Regions:

N/A

Comments:

N/A

Report Signed By

Name	Role	Date

Biomarker Descriptions

FLT3 (fms related receptor tyrosine kinase 3)

Background: The FLT3 gene encodes the fms related tyrosine kinase 3, a tyrosine kinase receptor that is a member of the class III receptor tyrosine kinase family that also includes PDGFR, FMS, and KIT[1]. FLT3 is highly expressed in hematopoietic progenitor cells[2]. Genomic alterations in FLT3 activate downstream oncogenic pathways including PI3K/AKT/mTOR and RAS/RAF/MEK/ERK pathways which promote cellular proliferation, survival, and inhibition of differentiation[1].

Alterations and prevalence: Somatic mutations occur in approximately 30% of acute myeloid leukemia (AML), 7-10% of melanoma, and up to 8% of uterine cancer[3,4,5,6]. The most common activating FLT3 mutations are internal tandem duplications (ITD) that range from 3 to 400 base pairs in length within exons 14 and 15 in the juxtamembrane (JM) domain[7]. The second most frequent mutations are point mutations in exon 20 within the tyrosine kinase domain (TKD)[8]. FLT3 is amplified in up to 8% of colorectal cancer, 3% of stomach cancer, and is commonly overexpressed in AML[5,6,9].

Potential relevance: The presence of FLT3-ITD confers poor prognosis in myelodysplastic syndrome (MDS)[10]. Concurrent expression of FLT-ITD with mutant or wild-type NPM1 (when lacking adverse risk genetic lesions) confers intermediate risk in AML[11,12]. FLT3 TKD mutation at D835 confers poor prognosis in MDS[10]. Midostaurin[13] (2017) and gilteritinib[14] (2018) are kinase inhibitors approved for AML patients with FLT3-ITD and TKD mutations including D835 and I836 mutations. The FDA granted fast track designations in 2017 to crenolanib[15] and in 2022 to tuspetinib (HM43239)[16] for FLT3 mutation-positive relapsed or refractory AML. In 2018 the FDA granted breakthrough therapy designation to quizartinib[17] for AML with FLT3-ITD. A phase II trial testing crenolanib in 34 patients with FLT3-ITD and TKD mutated relapsed/refractory AML, reported that FLT3 inhibitor naïve patients demonstrated a longer overall survival (OS) and event free survival (EFS) in comparison to previously treated patients (median OS: 55 weeks vs 13 weeks; median EFS: 13 weeks vs 7 weeks)[18]. Another phase II trial of crenolanib with chemotherapy in newly diagnosed FLT3 mutated AML reported complete remission in 24/29 (83%) patients[19]. Several multi-targeted tyrosine kinase inhibitors such as sorafenib (2005), sunitinib (2006), cabozantinib (2012), and ponatinib (2012) are FDA approved and include FLT3 as a target. Sorafenib is recommended in combination with chemotherapy in FLT3-ITD mutated AML[11].

NUP98 (nucleoporin 98 and 96 precursor)

Background: The NUP98 gene encodes the nucleoporin 98 and 96 precursor, a 186 kDa protein that undergoes autoproteolytic cleavage to generate one 98 kDa and one 96kDa nucleoporin (Nup98 and Nup96)[20,21]. Both Nup98 and Nup96 are components of the nuclear pore complex (NPC) which regulates the transport of molecules between the nucleus and cytoplasm[20,21,22]. Nup98 functions in several cellular processes including nuclear import and export through interaction with karyopherin α, a protein of the NPC which is responsible for the recognition of nuclear localization signals (NLS) and nuclear export signals (NES)[20,21]. Although less is known about the function of Nup96, it has been observed to interact with SEC13, a component of both COPII vesicles in the endoplasmic reticulum as well as the NUP complex[22,23,24]. The NUP98 gene is commonly observed to be fused with other partner genes in hematological malignancies, which has been shown to result in impaired hematopoietic precursor differentiation and increased stem or progenitor cell self-renewal[25].

Alterations and prevalence: Fusion of NUP98 was first identified in acute myeloid leukemia (AML) with the t(7;11)(p15;p15) translocation[26]. This translocation results in the fusion of the amino-terminal of NUP98 to the carboxyl-terminal of HOX9A[25,26]. NUP98 fusion with at least 28 different partner genes have been identified in several additional hematological malignancies including chronic myeloid leukemia in blast crisis (BP-CML), myelodysplastic syndrome (MDS), acute lymphoblastic leukemia (ALL), and bilineage/biphenotypic leukemia[25]. Fusions and rearrangements of NUP98 are observed in 2% of AML and less than 1% of solid tumors including sarcoma, bladder urothelial carcinoma, prostate adenocarcinoma, uterine corpus endometrial carcinoma, breast invasive carcinoma, and lung adenocarcinoma[5,6]. Somatic mutations in NUP98 are observed in 9% of uterine corpus endometrial carcinoma, 8% skin cutaneous melanoma, 5% of stomach adenocarcinoma, 4% of colorectal adenocarcinoma, 3% of lung adenocarcinoma, bladder urothelial carcinoma, cholangiocarcinoma, and lung squamous cell carcinoma, and 2% of diffuse large B-cell lymphoma, cervical squamous cell carcinoma, and esophageal adenocarcinoma.[5,6]. Biallelic deletion of NUP98 is also observed in 4% of brain lower grade glioma[5,6].

Potential relevance: Currently, no therapies are approved for NUP98 aberrations.

Relevant Therapy Summary

● In this cancer type ○ In other cancer type ◐ In this cancer type and other cancer types ✖ No evidence

FLT3 ITD mutation

Relevant Therapy	FDA	NCCN	Clinical Trials*
gilteritinib	●	●	✖
midostaurin + cytarabine + daunorubicin	●	●	✖
azacitidine	✖	●	✖
cytarabine + daunorubicin	✖	●	✖
cytarabine + daunorubicin + etoposide	✖	●	✖
cytarabine + etoposide + idarubicin	✖	●	✖
cytarabine + fludarabine + idarubicin + filgrastim	✖	●	✖
cytarabine + idarubicin	✖	●	✖
cytarabine + mitoxantrone	✖	●	✖
decitabine	✖	●	✖
midostaurin + cytarabine	✖	●	✖
sorafenib	✖	●	✖
sorafenib + azacitidine	✖	●	✖
sorafenib + decitabine	✖	●	✖
venetoclax + azacitidine	✖	●	✖
venetoclax + cytarabine	✖	●	✖
venetoclax + decitabine	✖	●	✖
chemotherapy, midostaurin	✖	✖	● (III)
crenolanib, chemotherapy	✖	✖	● (III)
crenolanib, midostaurin, chemotherapy	✖	✖	● (III)
gemtuzumab ozogamicin, chemotherapy, gilteritinib	✖	✖	● (III)
SKLB1028	✖	✖	● (III)
chemotherapy	✖	✖	● (II/III)
chemotherapy, venetoclax	✖	✖	● (II/III)
ibrutinib, sorafenib, chemotherapy	✖	✖	● (II/III)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Relevant Therapy Summary (continued)

● In this cancer type ○ In other cancer type ◐ In this cancer type and other cancer types ✖ No evidence

FLT3 ITD mutation (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
chemotherapy, gilteritinib, midostaurin	✖	✖	● (II)
chemotherapy, radiation therapy	✖	✖	● (II)
HYML-122	✖	✖	● (II)
HYML-122, chemotherapy	✖	✖	● (II)
radiation therapy, allogeneic stem cells, chemotherapy	✖	✖	● (II)
CC-90009, gilteritinib	✖	✖	● (I/II)
gilteritinib, chemotherapy	✖	✖	● (I/II)
gilteritinib, venetoclax, chemotherapy	✖	✖	● (I/II)
KDS-1001, cytokine, tacrolimus, mycophenolate mofetil, radiation therapy, chemotherapy, allogeneic stem cells	✖	✖	● (I/II)
midostaurin, gemtuzumab ozogamicin, chemotherapy	✖	✖	● (I/II)
NMS-P948	✖	✖	● (I/II)
quizartinib, chemotherapy	✖	✖	● (I/II)
stem cell therapy	✖	✖	● (I/II)
venetoclax, gilteritinib, chemotherapy	✖	✖	● (I/II)
allogeneic double negative T cells	✖	✖	● (I)
clifutinib	✖	✖	● (I)
gemtuzumab ozogamicin, midostaurin, chemotherapy	✖	✖	● (I)
iadademstat, gilteritinib	✖	✖	● (I)
LNK01002	✖	✖	● (I)
nintedanib, chemotherapy	✖	✖	● (I)
PHI-101	✖	✖	● (I)
venetoclax, chemotherapy	✖	✖	● (I)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Relevant Therapy Summary (continued)

● In this cancer type ○ In other cancer type ◑ In this cancer type and other cancer types ✖ No evidence

NUP98-NSD1 fusion

Relevant Therapy	FDA	NCCN	Clinical Trials*
radiation therapy, allogeneic stem cells, chemotherapy	✖	✖	● (II)
revumenib	✖	✖	● (I/II)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Clinical Trials Summary

FLT3 ITD mutation

NCT ID	Title	Phase
NCT04174612	A Phase III, Prospective, Randomized Multi-center Intervention Trial of Early Intensification in AML Patients Bearing FLT3 Mutations Based on Peripheral Blast Clearance: A MYNERVA-GIMEMA Study	III
NCT03250338	Phase III Randomized, Double-blind, Placebo-controlled Study Investigating the Efficacy of the Addition of Crenolanib to Salvage Chemotherapy Versus Salvage Chemotherapy Alone in Subjects < or = 75 Years of Age With Relapsed/Refractory FLT3 Mutated Acute Myeloid Leukemia	III
NCT03258931	Phase III Randomized Study of Crenolanib Versus Midostaurin Administered Following Induction Chemotherapy and Consolidation Therapy in Newly Diagnosed Subjects With FLT3 Mutated Acute Myeloid Leukemia	III
NCT04293562	A Phase III Randomized Trial for Patients With De Novo AML Comparing Standard Therapy Including Gemtuzumab Ozogamicin (GO) to CPX-351 With GO, and the Addition of the FLT3 Inhibitor Gilteritinib for Patients With FLT3 Mutations	III
NCT04716114	A Phase III, Open-Label, Multicenter, Randomized Study of SKLB1028 Versus Salvage Chemotherapy in Patients With FLT3-mutated Acute Myeloid Leukemia Refractory to or Relapsed After First-line Treatment(ALIVE)	III
NCT03256071	Low Dose Decitabine + Modified BUCY Conditioning Regimen for High Risk Acute Myeloid Leukemia Undergoing Allo-HSCT	II/III
NCT02416388	Phase II/III Randomized Study to Improve Overall Survival in 18 to 60 Year-old Patients, Comparing Daunorubicin Versus High Dose Idarubicin Induction Regimens, High Dose Versus Intermediate Dose Cytarabine Consolidation Regimens, and Standard Versus Mycophenolate Mofetil Prophylaxis of Graft Versus Host Disease in Allografted Patients in First CR : a Backbone InterGroup-1 Trial	II/III
NCT03642236	Combination of Brutons Tyrosine Kinase (BTK) Inhibitor Overcomes Drug-resistance in Refractory/ Relapsed FLT3 Mutant Acute Myeloid Leukemia (AML)	II/III
NCT02115295	Phase II Study of Cladribine Plus Idarubicin Plus Cytarabine (ARAC) in Patients With AML, HR MDS, or Myeloid Blast Phase of CML	II
NCT03121014	A Phase II Study of Intensity Modulated Total Marrow Irradiation (IM-TMI) in Addition to Fludarabine/ Busulfan Conditioning for Allogeneic Transplantation in High Risk AML and Myelodysplastic Syndromes	II

Clinical Trials Summary (continued)

FLT3 ITD mutation (continued)

NCT ID	Title	Phase
NCT05241106	A Single-arm, Open, Multicenter, Phase II Study to Investigator the Efficacy and Safety of HYML-122 in Patients With FLT3 Positive Relapsed or Refractory Acute Myeloid Leukemia (AML)	II
NCT05241093	A Single-arm, Open, Multicenter, Phase II Study to Investigator the Efficacy and Safety of HYML-122 and Cytarabine in Patients With FLT3 Positive Relapsed or Refractory Acute Myeloid Leukemia (AML)	II
No NCT ID	Phase II Study Of Allogeneic Hematopoietic Cell Transplantation For Children With Acute Myeloid Leukemia In First And Second Complete Remission Using Fludarabine, Cytarabine, Melphalan And Low-Dose Total Body Irradiation As A Conditioning Regimen (AML-SCT15).	II
NCT04336982	An Exploratory Phase 1b Open-label Multi-arm Trial to Evaluate the Safety and Efficacy of CC-90009 in Combination With Anti-Leukemia Agents in Subjects With Acute Myeloid Leukemia	I/II
NCT04240002	A Phase I/II, Multicenter, Open-Label, Single Arm, Dose Escalation and Expansion Study of Gilteritinib (ASP2215) Combined With Chemotherapy in Children, Adolescents and Young Adults With FMS-like Tyrosine Kinase 3 (FLT3)/Internal Tandem Duplication (ITD) Positive Relapsed or Refractory Acute Myeloid Leukemia (AML)	I/II
NCT05520567	A Phase I/II, Multicenter, Open-Label, Randomized Dose Ranging and Expansion Study of the Combination of Gilteritinib, Venetoclax and Azacitidine in Patients With Newly Diagnosed FLT3 Mutated Acute Myeloid Leukemia (AML) Not Eligible for Intensive Induction Chemotherapy	I/II
NCT05115630	A Phase I/II Clinical Trial of "Off-the-shelf" NK Cell Administration in Combination With Allogeneic SCT to Decrease Disease Relapse in Patients With High-risk Myeloid Malignancies Undergoing Matched Related, Matched Unrelated, One Antigen Mismatched Unrelated, or Haploidentical Stem-cell Transplantation	I/II
NCT04385290	MidOStaurin + Gemtuzumab OzogAmIcin Combination in First-line Standard Therapy for Acute Myeloid Leukemia (MOSAIC).	I/II
NCT03922100	A Phase I/II Study of NMS-03592088, a FLT3, KIT and CSF1R Inhibitor, in Patients With Relapsed or Refractory AML or CMML	I/II
NCT03793478	A Phase I/II, Multicenter, Dose-Escalating Study To Evaluate the Safety, Pharmacokinetics, Pharmacodynamics, and Efficacy Of Quizartinib Administered in Combination with Re-Induction Chemotherapy, and as a Single-Agent Continuation Therapy, in Pediatric Relapsed/Refractory AML Subjects Aged 1 Month to <18 Years (and Young Adults Aged up to 21 Years) with FLT3-ITD mutations.	I/II
NCT01203722	Reduced Intensity, Partially HLA Mismatched Allogeneic BMT for Hematologic Malignancies Using Donors Other Than First-degree Relatives	I/II
NCT05010122	A Phase I/II Study of ASTX727, Venetoclax, and Gilteritinib for Patients With Acute Myeloid Leukemia or High-Risk Myelodysplastic Syndrome With an Activating FLT3 Mutation	I/II
No NCT ID	Safety and Effective of Prophylactic Infusion of DNT cells in Allogeneic Hematopoietic Stem Cell Transplantation for Patients with Myeloid Malignancies: A Single Center Clinical Research	I
No NCT ID	A Phase I Study to Assess the Safety of Micro-dose Lenalidomide as Maintenance Therapy Post-allogeneic Haematopoietic Cell Transplantation for Patients with Acute Myeloid Leukaemia or Myelodysplastic Syndromes, at High Risk of Relapse	I

Clinical Trials Summary (continued)

FLT3 ITD mutation (continued)

NCT ID	Title	Phase
NCT04827069	A Phase I, Multi-center, Open,Single Arm, Dose-escalation Study to Evaluate the Safety, Tolerability, Pharmacokinetics and Efficacy of Clifutinib Besylate(HEC73543) in Relapsed or Refractory Acute Myeloid Leukemia (AML)	I
NCT03900949	A Phase I Study to Evaluate the Safety and Tolerability of Gemtuzumab Ozogamicin and Midostaurin When Used in Combination With Standard Cytarabine and Daunorubicin Induction for Newly Diagnosed FLT3-mutated Acute Myeloid Leukemia	I
NCT05546580	An Escalation/Expansion, Open Label, Multicenter Study of Iadademstat and Gilteritinib in Patients With Relapsed or Refractory Acute Myeloid Leukemia (R/R AML) With FMS-like Tyrosine Kinase Mutation (FLT3 Mut+): The FRIDA Study	I
NCT04896112	An Open-Label, Multicenter, Phase I Study to Evaluate the Safety, Pharmacokinetics and Preliminary Efficacy of LNK01002 in Patients With Malignant Myeloid Hematologic Neoplasms	I
NCT03513484	Phase Ib, Open Label, Combination Study of Nintedanib With 5-Azacitidine in Acute Myeloid Leukemia Characterized by HOX Gene Overexpression, That Are Not Candidates of Intensive Chemotherapy	I
NCT04842370	A Prospective, Phase Ia/Ib, Multicenter Study to Evaluate the Safety, Tolerability, Pharmacokinetics and Pharmacodynamics of the FLT3 Inhibitor, PHI 101, Alone in Subjects With Relapsed or Refractory Acute Myeloid Leukemia (AML).	I
NCT03613532	A Phase I Study of Adding Venetoclax to a Reduced Intensity Conditioning Regimen and to Maintenance in Combination With a Hypomethylating Agent After Allogeneic Hematopoietic Cell Transplantation for Patients With High Risk AML, MDS, and MDS/MPN Overlap Syndromes	I

NUP98-NSD1 fusion

NCT ID	Title	Phase
No NCT ID	Phase II Study Of Allogeneic Hematopoietic Cell Transplantation For Children With Acute Myeloid Leukemia In First And Second Complete Remission Using Fludarabine, Cytarabine, Melphalan And Low-Dose Total Body Irradiation As A Conditioning Regimen (AML-SCT15).	II
NCT04065399	AUGMENT-101: A Phase I/II, Open-label, Dose-Escalation and Dose-Expansion Cohort Study of SNDX 5613 in Patients With Relapsed/Refractory Leukemias, Including Those Harboring an MLL/KMT2A Gene Rearrangement or Nucleophosmin 1 (NPM1) Mutation	I/II

Assay Information

Methodology, Scope, and Application:The NCI Myeloid Assay v2 (NMAv2) is a next-generation sequencing (NGS) assay which identifies pre-defined and novel genomic variants covering 45 DNA genes and 35 fusion drivers that are categorized into 3 mutation types: single nucleotide variants (SNVs), insertions/deletions (Indels) including FLT3 internal tandem duplications (ITDs), and gene fusions. The assay utilizes Thermo Fisher Scientific's Oncomine® Myeloid Assay GX v2, a NGS assay that utilizes a multiplex polymerase chain reaction (PCR) with DNA and RNA extracted from blood and bone marrow mononuclear cells for sequencing on the Ion Torrent Genexus Integrated Sequencer and analyzed by the Ion Torrent Genexus pipeline version 6.6.2.1. The NMAv2 can reliably identify the presence or absence of 1661 hotspot variants and 779 reported fusions compared to the Human Reference Genome hg19, including the genes listed below. The NMAv2 is a laboratory developed test designed to detect gene mutations for major myeloid disorders.

Analytical Sensitivity and Specificity:The NMAv2 has been determined to be suitably analytically sensitive and specific for the various types of abnormalities within its reportable range, demonstrating 98.97% sensitivity, 100% specificity and an overall reproducibility with a mean positive percent agreement of 99% and a mean negative percent agreement of 100% during validation of the assay.

Hereditary and Germline Mutations:This assay examines cancer cells only and does not examine normal (non-cancer) cells. Mutations detected by the assay may be present only in the cancer cells, or in every cell of the body (including non-cancer cells). This test also cannot tell whether a potential germline mutation causes or will cause a hereditary cancer syndrome. If the patient's history includes any one or more features suggestive of a possible hereditary cancer predisposition (such as, cancer arising at a young age, especially a cancer of a type unusual in younger patients; a personal history of multiple different types of cancers; or a significant cancer history among blood relatives, especially but not exclusively of the same type as the patient's), it is recommended that the physician arrange for the patient to meet with a genetic counselor and, if warranted, undergo the appropriate genetic test on normal (i.e. non-cancer) tissue (blood or cells brushed from the oral surface of the cheek) to check for a germline abnormality, regardless of the results of this research study.

Report Content:Thermo Fisher Scientific's Ion Torrent Oncomine Knowledgebase Reporter (OKR) software was used in the generation of this report. Software was developed and designed internally by Thermo Fisher Scientific. The analysis was based on the current version of OKR. The data presented here is from a curated knowledgebase of publicly available information but may not be exhaustive. FDA information was sourced from www.fda.gov and is current as of this version. NCCN information was sourced from www.nccn.org and reflects its current version. Clinical Trials information reflects the current version. For the most up-to-date information regarding particular trials, search www.clinicaltrials.gov by NCT ID or search local clinical trials authority website by local identifier listed in 'Other Identifiers.' Variants are reported according to HGVS nomenclature and classified following AMP/ASCO/CAP guidelines (Li et al. 2017). Based on the data sources selected, variants, therapies, and trials identified in this report are listed in order of potential clinical significance but not for predicted efficacy of the therapies.

Genes assayed for hotspot mutations:
ABL1, ANKRD26, BRAF, CBL, CSF3R, DDX41, DNMT3A, FLT3, GATA2, HRAS, IDH1, IDH2, JAK2, KIT, KRAS, MPL, MYD88, NPM1, NRAS, PPM1D, PTPN11, SETBP1, SF3B1, SMC1A, SMC3, SRSF2, U2AF1, WT1

Genes assayed with full exon coverage:
ASXL1, BCOR, CALR, CEBPA, ETV6, EZH2, IKZF1, NF1, PHF6, PRPF8, RB1, RUNX1, SH2B3, STAG2, TET2, TP53, ZRSR2

Genes assayed for detection of fusions:
ABL1, ABL2, BCL2, BRAF, CCND1, CREBBP, EGFR, ETV6, FGFR1, FGFR2, FUS, HMGA2, JAK2, KAT6A, KAT6B, KMT2A, KMT2A-PTD (Partial Tandem Duplication), MECOM, MET, MLLT10, MRTFA (MKL1), MYBL1, MYH11, NTRK2, NTRK3, NUP214, NUP98, PAX5, PDGFRA, PDGFRB, RARA, RUNX1, TCF3, TFE3, ZNF384

Limitaions of the assay

1. Mutations in homopolymer regions of >5 nucleotides may not be accurately assessed by this assay.
2. FLT3-ITDs greater than 120bps may not be detected by this assay. Sensitivity for larger FLT3-ITD's may be reduced and variant allele frequencies may be reported lower than expected.
3. Following genetic alterations will not be reported by this assay: RUNX1 p.Ser445AlafsTer149, ASXL1 p.G646Wfs*12, PRPF8 p.Arg156GlufsTer28

DISCLAIMER

This assay reports SNV, indel and FLT3-ITD mutations ≥ 5% Variant Allele Frequency (VAF). This assay is considered a Laboratory Developed Test (LDT). Its performance characteristics have been determined through extensive testing although it has not been cleared by the US Food and Drug Administration and such approval is not required for clinical implementation. Furthermore, any comments included in the report are strictly interpretive and the opinion of the reviewer. This laboratory is certified under the Clinical Laboratory Improvements Amendments (CLIA) for the performance of high-complexity testing for clinical purposes. The correlative data presented is a result of the curation of published data sources but may not be exhaustive. The content of this report has not been evaluated or approved by the FDA or other regulatory agencies.

References

1. Grafone et al. An overview on the role of FLT3-tyrosine kinase receptor in acute myeloid leukemia: biology and treatment. Oncol Rev. 2012 Mar 5;6(1):e8. PMID: 25992210
2. Short et al. Emerging treatment paradigms with FLT3 inhibitors in acute myeloid leukemia. Ther Adv Hematol. 2019; 10: 2040620719827310. PMID: 30800259
3. Small. FLT3 mutations: biology and treatment. Hematology Am Soc Hematol Educ Program. 2006:178-84. PMID: 17124058
4. Cancer Genome Atlas Research Network. Genomic and epigenomic landscapes of adult de novo acute myeloid leukemia. N Engl J Med. 2013 May 30;368(22):2059-74. doi: 10.1056/NEJMoa1301689. Epub 2013 May 1. PMID: 23634996
5. Cerami et al. The cBio cancer genomics portal: an open platform for exploring multidimensional cancer genomics data. Cancer Discov. 2012 May;2(5):401-4. PMID: 22588877
6. Weinstein et al. The Cancer Genome Atlas Pan-Cancer analysis project. Nat. Genet. 2013 Oct;45(10):1113-20. PMID: 24071849
7. Nakao et al. Internal tandem duplication of the flt3 gene found in acute myeloid leukemia. Leukemia. 1996 Dec;10(12):1911-8. PMID: 8946930
8. Yamamoto et al. Activating mutation of D835 within the activation loop of FLT3 in human hematologic malignancies. Blood. 2001 Apr 15;97(8):2434-9. PMID: 11290608
9. Carow et al. Expression of the hematopoietic growth factor receptor FLT3 (STK-1/Flk2) in human leukemias. Blood. 1996 Feb 1;87(3):1089-96. PMID: 8562934
10. NCCN Guidelines® - NCCN-Myelodysplastic Syndromes [Version 1.2023]
11. NCCN Guidelines® - NCCN-Acute Myeloid Leukemia [Version 3.2022]
12. Döhner et al. Diagnosis and management of AML in adults: 2022 recommendations from an international expert panel on behalf of the ELN. Blood. 2022 Sep 22;140(12):1345-1377. PMID: 35797463
13. https://www.accessdata.fda.gov/drugsatfda_docs/label/2021/207997s008lbledt.pdf
14. https://www.accessdata.fda.gov/drugsatfda_docs/label/2022/211349s003lbl.pdf
15. https://www.globenewswire.com/news-release/2017/12/01/1216122/0/en/Arog-Pharmaceuticals-Receives-FDA-Fast-Track-Designation-for-Crenolanib-in-Relapsed-or-Refractory-FLT3-Positive-AML.html
16. https://www.aptose.com/news-media/press-releases/detail/230/aptose-receives-fast-track-designation-for-hm43239-in
17. https://www.daiichisankyo.com/media_investors/media_relations/press_releases/detail/006896.html
18. Jasleen et al. Results of a Phase II Study of Crenolanib in Relapsed/Refractory Acute Myeloid Leukemia Patients (Pts) with Activating FLT3 Mutations. Blood. 124:389
19. Bartus et al. Crenolanib in combination with standard chemotherapy in newly diagnosed FLT3-mutated AML patients. ASH 2017. Abstract #566
20. O'Leary et al. Reference sequence (RefSeq) database at NCBI: current status, taxonomic expansion, and functional annotation. Nucleic Acids Res. 2016 Jan 4;44(D1):D733-45. PMID: 26553804
21. Lam et al. NUP98 gene fusions in hematologic malignancies. Leukemia. 2001 Nov;15(11):1689-95. PMID: 11681408
22. Enninga et al. Sec13 shuttles between the nucleus and the cytoplasm and stably interacts with Nup96 at the nuclear pore complex. Mol Cell Biol. 2003 Oct;23(20):7271-84. PMID: 14517296
23. Faria et al. The nucleoporin Nup96 is required for proper expression of interferon-regulated proteins and functions. Immunity. 2006 Mar;24(3):295-304. PMID: 16546098
24. Bickford et al. A structural view of the COPII vesicle coat. Curr Opin Struct Biol. 2004 Apr;14(2):147-53. PMID: 15093828
25. Gough et al. NUP98 gene fusions and hematopoietic malignancies: common themes and new biologic insights. Blood. 2011 Dec 8;118(24):6247-57. PMID: 21948299
26. Borrow et al. The t(7;11)(p15;p15) translocation in acute myeloid leukaemia fuses the genes for nucleoporin NUP98 and class I homeoprotein HOXA9. Nat Genet. 1996 Feb;12(2):159-67. PMID: 8563754